LightOS

Powered By Autometra Technologies

**"Building the Future of Smart Lighting"**

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# Team behind the Innovation

## 1. Executive Team

| Member | Focus Area | Key Expertise & Contribution |
|---|---|---|
| Keshab Sharma | Team Lead, Vision, Communication & IoT | Expertise exists in firmware development. ESP32 microcontrollers also fall under this. Low-power circuit design is another specialty, drives the core LCU device intelligence. |
| Sidhu Kumar Singh | Business Management & Market Strategy | Focus is given to market penetration. Customer acquisition receives considerable effort. Strategic partnerships represent another major objective. We look forward to the changes he may be implementing. |
| Pratik Kumar Sah | Full Stack & Backend Business Logic | System Design, review software support, optimizes system performance and database efficiency. |
| Ravi Kant Thakur | Media Marketing & Graphics and Design | Ensures all the Design and graphics work, supply reliable design, and provide marketing assistance through social media. |

## 2. Advisory Mentors

| Member | Expertise & Guidance | Focus Area for LightOS |
|---|---|---|
| Dr. Raghvendra Kumar | Patent & Grant Support | Effectively guides the LightOS team. He assists us through patent filing. Grant application processes are also part of his role. He offers insight into product development. Also, he helps with field deployment strategies. His experience bridges the gap, innovation meets practical needs. |
| Dr. Sasank Sekhar Panda | Academic & Research Mentorship | His sharp analysis has been key. It shapes the team's direction, fostering team growth. He understands both technology and structure. He offers strong, insightful advice. It boost the very core foundation of the project, and makes it more viable for long-term viability. He manages important documentation. His continuous advice motivates us and It has driven project momentum. |
| Mr. Ranjit Patnaik | Hardware Support & Early Prototyping | Initial hardware development and testing involved several steps. Early prototypes emerged of the LightOS system, His technical input mattered. His hardware resources provided support for everyone. |

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# 1. The Problem:

## 1.1 Problem Overview:

Our targeted problem is the traditional public lighting system that operates across the country. In 2015 the government of India launched an initiative for the public light infrastructure, **Street Lighting National Program (SLNP)**. The program aimed to replace **Conventional Street light** with **Energy Efficient** and **Smart LED light** all across the nation. In addition, solar street lights are also being deployed in poor-connectivity regions, even though after all these initiatives, the ground-level reality still faces major inefficiencies in operation, monitoring and maintenances

## 1.2 **Key Problems and Issues:**

**1. Manual Intervention and Irregular Inspection:**

- Streetlights are still operated manually which requires human resources therefore increasing the operational costs.
- Due to lack of discipline or accountability, street lights often function at wrong time, delayed at night operation time or may left active during the day.
- Field Observation in rural areas shows that over 70% of streetlights function irregularly, which means either running during daytime or remaining OFF at night.

- Handling thousands of streetlights, with timely activation and deactivation seems challenging without proper budget allocation and manpower, causes poor maintenances, month long outages and raises security and safety concerns.
- Due to irregular or 24-hour operation of lights, the life expectancy decreases by up to 60%, increasing both power consumption and frequent replacements costs.

## 1.3 **Existing Infrastructure and Challenges**

1. **Control Switches:**
   - Conventional control switches require an extra parallel cabling, which is not feasible under modern single line distribution plans of different, state electricity distribution companies.
2. **LDR-Based Systems:**
   - Light Dependent Resistors (LDRs) are not that reliable when it comes to tracking the light failures, and differentiating between natural day light and artificial light conditions.
3. **Smart City-Focused Initiative:**
   - Existing smart streetlight solutions are particularly focusing on smart cities, as integration or replacement of traditional infrastructure complex and economically unviable for rural and semi-rural areas.

4. **Solar Street Light Challenges:**
    - Although the solar street light initiatives are successfully, but the post deployment maintenance is neglected by the authorities as well as citizens
    - Many tenders are for poor-connected regions like rural and semi-rural and requires mostly new pole, solar panel, fixtures, so the question arises what after the tender completed. *Who will be taking care of the lights if they are non-functional after 1 or 2 years?*

## 2. Our Vision:

Our Company **Autometra Technologies,** with product **LightOS** aims to bridge the gap between **traditional infrastructure and modern smart light control** by developing solutions that:

- Support both **rural and urban development** through scalable infrastructural automation.
- Transform **India's Public Lighting Infrastructure** into a global example of revolutionary innovation and optimization.
- Enable the **LightOS Dashboard** to control thousands of lights anytime, from anywhere.
- Give each public light a **unique identity over internet**, for easy monitoring and maintenance.
- Empower lights to become smart enough to **self-decision-making system**s, capable of detecting and reporting the issues to the central command center and requesting for maintenance.
- Promote **responsible utilization of the public resource**, higher output by reducing extensive power consumption, extending light lifespan and minimizing operational costs.
- Replace costly **imported Smart- Light solution** with our economical Indian solution.
- **Support for Digital India:** Foster smart infrastructure development that aligns with the Digital India initiative, enhancing the digital transformation of public utilities.
- **Vision for Viksit Bharat 2047:** Contribute towards the long-term vision of Viksit Bharat 2047, by promoting sustainable, tech-driven solutions that improve public infrastructure and quality of life across India.

# 3. The Solution:

The **LightOS** brings a economical and optimized approach for Indian public light infrastructural and development by incorporating advanced technologies such as **IoT, LoRa, Wi-Fi Mesh, Satellite Communication and Artificial Intelligence and Machine learning**.

## 3.1 Technological Overview

**a. Real-Time Automation:**

- Unlike traditional systems that rely on manual switching, LDR circuits or basic timers, LightOS includes real-time automation, utilizing sensors and programmable logic to adjust lighting based on environmental conditions.
- This ensures lights are active only when needed, drastically reducing unnecessary electricity consumption. The system can also adapt to local variations in sunrise andsunset times, offering a more dynamic and efficient solution than Traditional lightingsystems.

**b. Fault Detection and Reporting:**

- One of the impressive features of SPLS is its built-infault detection mechanism. The system continuously monitors the status of each streetlight and automatically identifies malfunctions such as burnt-out bulbs or wiring issues.
- This is communicated to a central system, reducing time of lightand late maintenance issues. Traditional street lighting often requires manual inspections, leading to delays in repairs and extended periods of darkness in public areas.

**c. Energy Efficiency through Advanced Control:**

- LightOS leverages energy-efficient technologies, including the use of LED lighting combined with automated control systems.
- By switching lights on and off based on real-time data, energy savings are maximized, directly reducing electricity bills.

**d. Scalability and Flexibility:**

- The system is designed to be easily scalable, capable ofbeing deployed across both rural and urban settings with minimal infrastructure changes.
- Its flexible architecture allows for the integration of various communication protocols, whether in areas with access to stable internet or in rural zones using mesh networks or local Wi-Fi solutions.

**e. Integration with Existing technology:**

- LightOS is easily compatible with the existing streetlight and can be easily installed across the operating streetlight. An enclosedevice will be provided that will be best suited for the running streetlight.

**f. Sustainability and Environmental Impact:**

- By reducing energy waste and optimizing streetlight operation, LightOS supports broader sustainability goals, contributing to thereduction of greenhouse gas emissions. Municipalities and rural Gram panchayats can benefit from significant cost savings and also promote eco-friendly practices.

## Solution Versioning:

The Solution versioning helps to meet and understand different stakeholders' requirements utilization and demands, the following versions/model meets different requirements:

| Version | Name | Connectivity | Description | Core Features | Stake Holders |
|---|---|---|---|---|---|
| v1.0 | LightOS Local | Offline (Wi-Fi /hotspot) | Standalone lighting control system using ESP webserver — no internet required. | Local scheduling, RTC-based automation, manual ON/OFF, local dashboard. | Rural areas, campuses, small colonies. |
| v2.0 | LightOS Remote | Cloud-enabled | Internet-connected version for remote scheduling and control. | Cloud dashboard (Node.js + MongoDB), real-time control, multi-user access, logs & analytics. | Municipalities, smart towns, institutions. |
| v3.0 | LightOS Smart | Cloud + Sensor-driven | Intelligent lighting with automation and data learning. | ML-based brightness control, fault alerts, predictive maintenance, energy analytics. | Smart cities, tech parks, gated societies. |
| v4.0 | LightOS Grid | Hybrid (Mesh + Solar) | Autonomous distributed system with solar power and mesh communication. | Solar-powered nodes, LoRa/Zigbee mesh, local fallback when offline, OTA updates. | City-wide grids, industrial zones. |
| v5.0 | LightOS One | Unified IoT Platform | Complete IoT ecosystem managing lighting and other utilities. | Integration with other city services (traffic, water), APIs, central command dashboard. | Governments, Smart City missions, global partners. |

# 4. The market potential

The Indian Smart Street Lighting Industry is undergoing a rapid digital transformation driven by the push for energy efficiency, automation, and sustainable infrastructure. According to industry estimates, the Indian smart street lighting market was valued at USD 58.3 million in 2024 and is projected to grow to USD 455 million by 2035, registering a CAGR of ~20.3% (2025–2035).

This market encompasses the hardware, software, connectivity, and integrated service ecosystem required for intelligent street lighting control — including remote monitoring, automated scheduling, real-time fault detection, and energy optimization.

## 4.1 Target Customer Segments

SPLS (Smart Public Lighting System) is designed to serve multiple high-potential verticals:

- Urban Local Bodies (ULBs) and Municipal Corporations under the *Smart Cities Mission*
- Government-led programs such as the *Street Lighting National Programme (SLNP)*
- Private infrastructure developers, industrial parks, special economic zones (SEZs), and gated communities
- Rural and hilly regions, where standalone or low-connectivity lighting automation is critical for safety and cost efficiency

## 4.2 Market Potential Analysis

India currently operates over 45+ million public street lights, with nearly 60% (≈27 million) identified as potential candidates for smart automation over the next decade.

**Hardware Market (One-time Installation)**

| | |
|---|---|
| **Estimated Adaptation:** | 21 million Units |
| **Average Device Cost:** | ₹600 -₹800 per unit (depending on model) |
| **Market potential:** | ₹12,600 crore and ₹16,800 crore. |

**Software & Services Market (Recurring Annual Revenue)**

| | |
|---|---|
| **Cloud and remote monitoring services:** | ₹20/per unit light monthly |
| **Monthly Market Potential:** | ₹42 crore |
| **Annual Recurring Market Potential (ARR):** | ₹504 crore |
| **Total Addressable Market (Hardware + Software):** | ₹13,100–₹17,300 crore (India) |

## 4.3 Key Industry Growth Drivers

**i. Government Initiatives & Policy Support**

- Programs such as SLNP (Street Lighting National Programme) and the Smart Cities Mission are accelerating the adoption of intelligent lighting systems.
- Public-Private Partnerships (PPP) and energy service company (ESCO) models are incentivizing municipalities to adopt smart automation solutions.

ii. **Energy Efficiency Targets**

- Street lighting accounts for 1–2% of India's total electricity use.
- Smart systems can reduce energy consumption by 50–70%, driving major cost savings for local bodies and DISCOMs.
- Integration with IoT-based energy analytics aligns with national sustainability goals.

iii. **Urbanization & Infrastructure Expansion**

- The rapid growth of tier-2 and tier-3 cities is creating demand for automated, centrally managed lighting systems across highways, industrial corridors, and public zones.
- Infrastructure modernization and ESG compliance (Environmental, Social, and Governance) are further accelerating smart lighting investments.

## 4.4 Potential Areas of Application in the Market

i. **Residential Communities:**

- It can be installed in gated communities and apartment complexes, ensuring security lighting with efficient energy use.

ii. **Rural Areas:**

- Can be implemented in rural panchayats and villages to improve public lighting with a cost-effective and scalable solution.

iii. **Public Institutions:**

- Applicable in educational campuses, government buildings, and hospitals, providing secure, well-lit environments while reducing operational costs.

iv. **Highway Lighting:**

- The long highways of India contain a high number of streetlights that needs to be regulated. LightOS can monitor and regulate the street light and automatically detect any fault.

v. **Urban Public Lighting:**

- Ideal for city streets, highways, and public parks where efficient lighting management is crucial for safety and visibility.

vi. **Industrial Complexes:**

- Useful in industrial zones, warehouses, and factories where automated and optimized lighting can reduce energy consumption and maintenance costs.

vii. **Smart Cities projects:**

- Fits seamlessly into smart city projects, where IoT solutions are used for improving infrastructure and urban services.

viii. **Border Areas:**

- It can help in knowing the status of streetlight deployed across border areas and may help in automated switching (on and off) at any time if any vulnerability caused.

# 5. Financial Projections and Return on Investment (ROI)

**Objective:** To demonstrate that LightOS is a self-funding investment that converts massive Operational Expenditure (OPEX) into measurable, recurring savings through **Dynamic Optimization** and **Proactive Fault Remediation (PFR)**.

## 5.1 Unit Economics & Core Assumptions

| Parameter | Value | Details |
|---|---|---|
| **Electricity Tariff (Odisha)** | **₹6.50 / kWh** (650 paisa) | Rate used for all cost calculations. |
| **Traditional Operation** | 12 hours/day | Standard assumed run-time for unmanaged lighting (e.g., 50W/80W). |
| **LightOS Optimized Lights** | **10W LED** running for **8 hours/day** | Your specified reduced load and operational time for the managed light. |
| **LightOS LCU Device Consumption** | **Active:** 10W (10 hours/day) /**Hibernation:** 3W (14 hours/day) | Power consumed by the Control Unit itself. |
| **LightOS LCU Subscription** | **₹240 / light / year** | Recurring OPEX for software, cloud, and PFR alerts. |

## 5.2 Annual Operating Cost: Traditional vs. LightOS

### A. Total Cost of Traditional System (Baseline)

The light runs unmanaged for 12 hours at full wattage.

| Light Wattage | Daily Consumption (12h) | Annual Energy Cost (≈ Total Cost) |
|---|---|---|
| **80W** | 960 Wh | ₹2,277.60 |
| **50W** | 600 Wh | ₹1,423.50 |

**B. Total Cost of LightOS Managed System (Optimized OPEX)**

This figure represents the **Total Annual Cost (B)** for running the LightOS solution, including the optimized light's energy, the LCU device's energy, and the subscription fee.

**i. LightOS Optimized Energy Consumption**

This calculates the energy cost for the **optimized 50W LED Light** running 8 hours, and the **LCU device** running 24 hours.

| Operational Components | LCU Active Mode (10W/10h) | LCU Hibernation Mode (3W/14h) | Daily Consumption (Wh) for Light (50W/8h) | Daily Consumption (Wh) for Light (80W/8h) |
|---|---|---|---|---|
| **Energy consumption (₹/ day)** | 100Wh x ₹6.50/1000= **₹0.65** | 42 Wh x ₹6.50/1000 =**₹0.273** | 400 Wh x ₹6.50/1000 =**₹2.60** | 640 Wh x ₹6.50/1000 =**₹4.16** |
| **Annual Energy Cost** | **₹237.35** | **₹99.64** | **₹949.00** | **₹1518.40** |

**ii. LightOS Grand Total Annual OPEX (B)**

| Cost Component | 50W Light | 80W Light |
|---|---|---|
| **LCU (Active + Hibernation)** | **₹336.99 + ₹949** | **₹336.99 + ₹1518.40** |
| **Annual Software Subscription** | ₹20 x 12 month = **₹240.00** | |
| **LightOS Total Annual OPEX (B)** | **₹1285.99** | **₹2095.39** |

## 5.3 Net Savings Per Light (The ROI)

The Net Savings is the direct difference between the **Traditional Cost (A)** and the **LightOS Total Annual OPEX (B)**.

| Light Wattage | Traditional Annual Cost (A) | LightOS Annual OPEX (B) | Net Annual Savings (A - B) | Effective OPEX Reduction |
|---|---|---|---|---|
| **50W** | ₹1,423.50 | ₹1285.99 | ₹137.51 | **9.66%** |
| **80W** | ₹2,277.60 | ₹2095.39 | ₹182.21 | **8.00%** |

### 5.4 Financial Impact Summary

- **Guaranteed Savings:** By migrating to LightOS, customers achieve a guaranteed annual OPEX reduction of **66.4%** on their highest-wattage lights, even after accounting for the subscription fee.
- **Self-Funding Investment:** The substantial annual savings ensures that the initial hardware CAPEX is recouped quickly, delivering a swift and compelling **Return on Investment (ROI)**.
- **Additional Savings:** This calculation does **not** include the reduction in labor and operational costs achieved through **Proactive Fault Remediation (PFR)**, which can add another 10-15% to total OPEX savings.

## 6. Business Model

### 6.1 Overview

Our company **Autometra Technologies** operates on a hybrid business model which combining **hardware sales** (CAPEX) and **LightOS Software** (SaaS) and **maintenance subscription** (OPEX). The business model ensures both immediate **capital gains** and **long-term** subscription support.

### 6.2 Revenue Model

| Stream | Description | Pricing | Frequency |
|---|---|---|---|
| Hardware (Device Sales) | Sale of LightOS Control Units (LCU) to municipalities, industries, and housing projects. | ₹1,500 – ₹1,800 per unit | One-time |
| Software & Cloud (SaaS) | LightOS Dashboard for monitoring, scheduling, and analytics. | ₹20 – ₹30 per light / month | Recurring |
| Maintenance Contracts (AMC) | Annual maintenance and system health check. | ≈ 10 % of hardware value | Yearly |
| Data & Analytics (Future) | Energy reports, predictive insights, and AI-based optimization. | Custom plan | Recurring |

### 6.3 Cost Model

The cost model is divided into CAPEX (Capital Expenditure) and OPEX (Operational Expenditure):

| Category | Description | Examples |
|---|---|---|
| CAPEX | One-time costs for hardware design, R&D, tooling, and initial setup. | PCB design, testing equipment, software licensing. |
| OPEX | Recurring costs for operations and services. | Cloud hosting, API gateway, maintenance staff, customer support |

Hardware revenue supports CAPEX recovery, while recurring SaaS and AMC subscriptions sustain OPEX, ensuring financial self-sufficiency beyond the first fiscal year.

## 6.4 Profit & Margin Summary

| Type | Avg. Cost | Selling Price/unit | Profit | Profit Margin |
|---|---|---|---|---|
| **LightOS Local (v1)** | ₹795 | ₹1200 | ₹405 | 50.9% |
| **LightOS Remote Standard (v2)** | ₹855 | ₹1300 | ₹445 | 52.0% |
| **LightOS Smart Advanced (v3)** | ₹1072 | ₹1600 | ₹528 | 49.3% |
| **LightOS Grid Pro (v4)** | ₹1297.5 | ₹2000 | ₹702.5 | 54.1% |
| **LightOS One Ecosystem Projects)** | ₹2880 | ₹4000 | ₹1120 | 38.9% |
| **Software (SaaS) (Monthly)** | ₹5 (up to v3)<br>₹8 (above v3) | ₹20–₹30<br>(Depending upon the version) | ₹18.5 (Avg.) | 85%-92% |
| **Annual Maintenance Cost (AMC)** | — | 10% of hardware value | | 20–40% |

## 6.5 Subscription Model

Every **1,000 lights** onboarded generates **₹2.4–₹3.0 lakh ARR**, providing predictable and scalable income. Once hardware is deployed, the recurring software revenue grows automatically without any additional CAPEX.

## 6.6 Financial Scaling Projection

| Year | Units Deployed | ARR (₹ lakhs) | Total Revenue (₹ lakhs) | Net Profit Margin |
|---|---|---|---|---|
| 2026 | | | | |
| 2027 | | | | |
| 2028 | | | | |

## 6.7 Go-to-Market Strategy

1. B2G (Business-to-Government): Smart City Mission, SLNP, and rural electrification projects.
2. B2B (Business-to-Business): Builders, industrial parks, and private campus projects.
3. Pilot-to-Scale Model: Begin with localized deployment (50–100 lights) → expand city-wide.
4. Channel Partners: Tie-ups with lighting manufacturers and contractors for faster scale.

## 6.8 Pricing Philosophy

1. Hardware: Minimum 3× cost multiplier to cover production, warranty, and R&D.
2. Software: Minimum 5× cost multiplier to cover hosting, data, and service support.
3. Aim for break-even within 18 months of initial deployment.

## 6.9 Value Proposition

1. Low-cost, Made-in-India alternative to imported smart lighting systems.
2. Offline + online operability ensures adoption across all terrains.
3. Recurring service model provides sustainable income for long-term scalability.

## 7. Current Traction

Summarized the core technical and IP milestones achieved to date, demonstrating execution and risk reduction.

| Milestone | Timeframe | Key Achievement | Risk Mitigated |
|---|---|---|---|
| 1. Hardware Feasibility Proven | Nov 2024 | Working Prototype (v1.0, v2.0), reached TRL (Technology readiness level) 4, Built and range tested, confirming all technical requirements and changes applied. | Technical & Feasibility Risk |
| 2. Product Viability Confirmed | Feb 2025 | Dashboard is fully functional, capable of real-time data ingestion. | Software & Integration Risk |
| 3. Core Technology Secured | April 2025 | Patent application initiated for our proprietary, Application ID: 202531040515 | IP (Intellectual Property) Risk |

## 8. Funding History and Investors

**The idea for LightOS was sparked by that very question: why can't street lighting be smart, efficient, and self-managing?**

- The thought for the change started with an innovative mind to counter one of the day to day to problem that we face but often neglect and it was came into effect with a team effort.
- With no external backing, except our mentors, at the beginning, we invested our own savings to turn a rough concept into a working prototype. That early risk - financial and emotional alike - served as the bedrock for what LightOS is now.
- Embracing the value of our early work, the Ministry of Micro, Small & Medium Enterprises (MSME), Government of India, recognized our innovation and partnered with us as a strategic funding partner to drive our development and deployment path.

### 8.1 Funding and Milestones:

| Year | Amount (INR) | Investor | Purpose |
|---|---|---|---|
| 2024 - 2025 | 15 thousand | Self-Funded | Concept validation and prototype development (hardware + firmware) |
| 2025 - 2026 | 6.9 Lakh | Govt. of India | Product refinement, infrastructure setup, and pilot launch |

# 9. Competitive Landscape

| Category | Competitor | Primary Focus | Market Scale | Key Strength | Identified Gap / Weakness |
|---|---|---|---|---|---|
| **Existing Players** | CITiLIGHT | Cloud-based smart street lighting systems for cities | Large scale (Metros, Smart Cities) | Proven deployments, IoT integration | No offline mode, high cost for small projects |
| | Neev Energy | Intelligent lighting with remote monitoring | Large municipal projects | Adaptive lighting, remote control | Limited modularity, no offline/local dashboard |
| | Havells/Bajaj /Syska | Hardware-led smart lighting | Nationwide | Manufacturing scale, brand trust | Weak in IoT software, no data ownership control |
| **Future competitors** | IoT Startups / Smart City Integrators | Low-cost IoT + sensor solutions | Small to medium | Rapid innovation, cloud-native | Lack of field experience, unreliable large-scale support |
| | Government Smart City Consortiums | Public smart city frameworks | Metro and large municipal | Policy support, funding access | Slow innovation cycles, not flexible for smaller projects |

# 10. Future Vision

As the system evolves towards Version 4, and then Version 5, LightOS will certainly integrate artificial intelligence. Real-time data analytics capabilities will be included too. This integration enables certain advantages:

## 10.1 AI Readiness

- **Predictive Energy Management:** Continuous monitoring of consumption helps predict usage patterns. It allows automatic optimization of brightness levels. These brightness adjustments are based on time, traffic and also the current weather conditions.
- **Proactive Maintenance:** AI models will analyze fault data. They do so to forecast potential failures. Downtime will be reduced, not to mention maintenance costs, but more can be done.
- **Anomaly Detection:** It can detect irregular power consumption, or outages very fast. This will, or so we expect, improve operational efficiency in every aspect.

## 10.2 Open Architecture & IoT Integration

LightOS is built, as it stands, on an open architecture; and a modular one. Allowing it seamless integration, we believe, with other smart infrastructure systems in time. Soon enough, the same LightOS-enabled poles should be able to act as multi-purpose IoT gateways, even supporting possibilities like:

- **Public Safety Integration:** Cameras and various sensors can now monitor pedestrian movement and traffic conditions; also important is monitoring public security in real-time which means quickly addressing risks to our community.

- **Environmental Monitoring:** Sensors will also integrate air quality and temperature metrics; and noise levels too in an attempt to provide, hopefully actionable, data on our collective impact. This is useful data, of course, for future city planning.
- **Smart City Expansion:** These IoT-equipped poles can be said to form the backbone. Certainly, they will for a broader smart infrastructure: they enable applications, say like, waste management. Furthermore, they help to establish smart parking areas and support of EV charging stations.

## 10.3 Business Perspective

The SPLS business model it is designed for both sustainability and for scalability as well. Future initiatives surely include, eventually:

- **On-Site Maintenance Teams:** Dedicated service units shall, in all likelihood, be deployed regionally so that the response becomes rapid. Not just responding to issues; a focus on preventing system failure, maintaining uptime and even system reliability will be included.
- **Data as a Service (DaaS):** The system will need scale, when scaling up; LightOS could collect precious data. Think, on energy usage; even data collected on lighting performance and of course urban activity! This data once made anonymous and aggregated, could actually then be sold; to city hall and agencies needing solid metrics upon which solid smart policy might exist in reality.

## 10.4 Sustainability & Environmental Contribution

LightOS can definitely support India's push, to its sustainability goals, it can assist their commitment to becoming a Green Country. All goals can be met.

- **Reduction in $CO_2$ Emissions:** Well-thought-out lighting schedules coupled; with well executed, effective dimming can potentially cut a good amount; off, in particular about 30-40% energy reduction, lowering our carbon imprint.
- **Cost effectiveness:** affordability to most places even really tiny towns or smaller places is our sincere hope; they need it the most really.